

ContractAI — Comprehensive Project Plan & Architecture

Author: Antigravity Engineering Team • **Target:** Offline Hackathon & Enterprise Student Intelligence Platform • **Version:** 2.4.0 (Production Release)

1. Executive Summary & Vision

ContractAI is a specialized student intelligence and contract analysis platform designed to demystify complex legal and institutional agreements for university students, interns, gig workers, and job applicants. By bridging dense legal language with accessible, student-centric insights, ContractAI protects individuals from exploitative lock-ins, undisclosed penalties, restrictive non-competes, and asymmetric termination clauses.

In addition to contract parsing, ContractAI embeds an intelligent **Student Contact Analyzer Engine** that categorizes, tracks, and analyzes relationship health across academic, professional, and peer networks. Built with dual-mode operational architecture, ContractAI executes seamlessly **100% offline with zero internet dependency** (using deterministic rule engines and localStorage persistence) while supporting cloud-scale expansion via FastAPI, MongoDB, and Gemini 1.5 Flash LLM APIs.

2. Problem Statement & Strategic Objectives

Core Challenge	Student Impact	ContractAI Solution
Dense Legal Jargon	Students routinely sign offer letters, PG leases, and bonds without recognizing severe obligations.	Extracts verbatim clauses and translates them into plain, actionable language categorized into 21 student topics.
Hidden Financial Penalties	Unnoticed forfeiture of security deposits, early exit damages, and IP forfeiture clauses.	Automated High/Medium/Low priority risk tagging with structured 'What You Need to Pay / Do / Avoid' matrices.
Network & Placement Tracking	Students lose track of high-priority faculty referrals, HR follow-ups, and recruiter deadlines.	Contact Analyzer with network health scoring, duplicate resolution, and inactive relationship alerts.
Demo & Hackathon Internet Instability	Cloud-only AI demos fail during offline jury evaluations or bandwidth-constrained hackathons.	100% deterministic offline engine, local SQLite/localStorage failover, and zero external CDN dependencies.

3. System Architecture & Tech Stack

ContractAI adopts a decoupled, modern client-server architecture engineered for high availability and offline resilience.

Layer	Technologies & Libraries	Key Architectural Responsibilities
Frontend Client	React 19, Vite 8, React Router v7, Tailwind CSS v4, Lucide React, Axios, Clsx	Single-Page Application (SPA), reactive state management, offline localStorage sync, responsive dark-mode UI, client-side route guards.
Backend API	Python 3.10+, FastAPI, Uvicorn, Pydantic v2, Python-Jose (JWT), Passlib (Bcrypt)	RESTful microservices, asynchronous lifespan handlers, bearer token authentication, request validation, CORS middleware.

Document Engine	PyMuPDF (fitz), Python-Multipart, ReportLab 5.0+	High-speed PDF text extraction with page preservation, layout reconstruction, dynamic multi-page PDF generation.
Intelligence Layer	Deterministic NLP Heuristic Engine + Google Gemini 1.5 Flash (Optional)	21-category pattern extraction, obligation synthesis, contextual Q&A; heuristics, offline failover orchestration.
Data Storage	MongoDB (Motor AsyncIO Driver) + In-Memory Store + Browser LocalStorage	Dual-mode hybrid persistence: Cloud/Local MongoDB database with seamless client-side and in-memory cache fallbacks.

4. Offline Resilience & Zero-Dependency Architecture

To guarantee 100% operational success during offline hackathon evaluations, pitch presentations, and disconnected campus environments, ContractAI employs a multi-tiered resilience model:

- **Zero External CDN Dependencies:** All stylesheet rules, fonts, and icon assets (Lucide-React) are bundled statically by Vite. No Google Fonts, unpkg, or external stylesheets are referenced.
- **Deterministic Local Rule Engine:** When GEMINI_API_KEY is unconfigured or no internet is detected, ContractAI's 21-category regex/NLP engine runs locally in <15ms without external requests.
- **Hybrid Persistence Pipeline:** If MongoDB is not running, the backend falls back to in-memory dictionaries, and the frontend falls back to browser localStorage databases (contractai_student_contacts, contractai_registered_users_db, contractai_offline_analyses).
- **One-Click Instant Demo Seeder:** The **Load Demo Data** feature populates 20 realistic student records across 7 distinct categories (Teachers, College, Friends, Internship, Placement, Hackathon, Professional) alongside pre-computed contract analyses.

5. Detailed Functional Module Breakdown

Module 1: Authentication Gate & Session Management

- **Route Protection:** Strict ProtectedRoute wrapping prevents unauthenticated access to dashboard, analyzer, results, and history.
- **Public Redirects:** PublicOnlyRoute automatically steers authenticated users from /login and /register directly to /dashboard.
- **Validation Rules:** Enforces email syntax, duplicate email rejection, minimum 6-character password matching, and session token persistence across page refreshes.
- **Default Credentials:** Pre-configured demo profile: student@university.edu / student123 with one-click autofill.

Module 2: Contact Intelligence & Network Analyzer

- **7-Category Taxonomy:** Organized across Teachers, College Admin, Friends/Peers, Internship Mentors, Campus Placement HRS, Hackathon Teammates, and Professional Advisors.
- **Network Health Metric:** Calculates composite score (0–100%) evaluating communication recency, high-priority touchpoints, and relationship coverage.
- **Risk & Inactivity Engine:** Highlights contacts neglected for >30 days and surfaces missing contact information (e.g. missing phone numbers).
- **Data Interoperability:** Full CSV Import parser with automated field mapping and one-click RFC4180 CSV export download.

Module 3: AI & Heuristic Contract Parsing Engine

- **21 Student Categories:** Evaluates Notice Periods, Stipends/Salaries, IP Assignment, Non-Compete Bounds, Confidentiality Fines, Lock-in Periods, Curfews, and Cancellation Rules.
- **Severity Classification:** Tags every finding as High, Medium, or Low attention priority based on student liability and monetary impact.
- **Obligations Matrix:** Synthesizes data into 'What You Must Pay', 'What You Must Do', 'What You Cannot Do', and 'What Happens If You Leave Early'.
- **Contextual Q&A; Chat:** In-page interactive assistant with clause citation snippets and intelligent suggested follow-up queries.

Module 4: Report Generation & Export Subsystem

- **PDF Document Builder:** Server-side ReportLab pipeline creating multi-page executive summaries, obligation breakdowns, and clause catalogs.
- **Offline Text Fallback:** Client-side generator providing structured .txt reports if backend PDF compilation is offline.
- **Printable View:** Browser-native print formatting with CSS page-break optimization for physical record keeping.

6. Database Schema & Data Models

ContractAI uses strict Pydantic schemas on FastAPI and schema-validated JSON document representations in MongoDB / localStorage:

Entity / Collection	Key Attributes & Types	Relationships & Indexes
User (users)	id: str name: str (2-100 chars) email: EmailStr (Unique) password_hash: str (Bcrypt) created_at: datetime	Primary auth identity. Unique index on email. 1-to-many relationship with Contracts & Analyses.
Contact (contacts)	id: str name: str, email: str phone: str, category: str tags: List[str] lastInteraction: str (YYYY-MM-DD) priority: 'High' 'Med' 'Low' interactionFrequency: str notes: str	Local & remote sync. Evaluated by Network Health and Inactivity algorithms. Categorized into 7 student groups.
Contract (contracts)	id: str, user_id: Optional[str] filename: str page_count: int full_text: str pages_data: List[{page_num, text}] uploaded_at: datetime	Parsed PDF document root. Child of User. Parent of AnalysisResult.
AnalysisResult (analyses)	id: str, contract_id: str summary: str, duration: str key_points: List[str] findings: List[FindingItem] student_obligations: ObligationsDict engine_used: 'gemini' 'heuristics' created_at: datetime	Output of AI/Heuristic pipeline. Referenced by Interactive Chat and PDF Report Builder.

7. REST API Specification & Endpoints

Method & Route	Auth Scope	Request / Response Payload
POST /api/auth/register	Public	Body: {name, email, password} Res: {access_token, token_type, user}
POST /api/auth/login	Public	Body: {email, password} Res: {access_token, token_type, user}
GET /api/auth/me	Bearer Token	Res: {id, name, email, created_at}
POST /api/contracts/upload	Optional Token	Form: file: Multipart PDF Res: ContractResponse {id, filename, page_count}
POST /api/contracts/upload-text	Optional Token	Body: {text, filename} Res: ContractResponse
POST /api/analysis/{contract_id}	Optional Token	Res: AnalysisResult (Summary, Findings, Obligations)
GET /api/analysis/{analysis_id}	Optional Token	Res: AnalysisResult
GET /api/analysis/history	Bearer Token	Res: List[AnalysisHistoryItem]
POST /api/analysis/{id}/chat	Optional Token	Body: {message, history} • Res: {reply, citations, followups}
GET /api/reports/{id}/pdf	Public / Token	Res: Binary Application/PDF Stream

8. User Journey & Navigation Sitemap

The application enforces a secure, distraction-free workflow with fluid glassmorphic UI patterns, high-contrast dark palette, and reactive feedback:

Step / Stage	Screen / Route	User Actions & Interaction Flow
1. Discovery	/ (Home)	View hero value prop, explore feature cards (AI Analysis, Contacts, Insights, Risks, Reports), click Get Started.
2. Authentication	/login & /register	Login with demo credentials (student@university.edu / student123) or register new account with instant validation.
3. Dashboard	/dashboard?tab=overview	Overview statistics, click 'Load Demo Data' button to seed 20 student contacts and pre-computed contract analyses.
4. Contact CRM	/dashboard?tab=contacts	Filter by 7 categories, search by name/tag, create new contacts, import/export CSV, view relationship health.
5. Insights & Risks	/dashboard?tab=insights	Review duplicate contacts, identify inactive professors/mentors (>30 days), inspect AI recommendations.
6. Contract Analysis	/analyzer	Upload PDF or paste agreement text. Click 'Analyze Contract' to trigger local or cloud AI extraction.
7. Deep Findings	/results/:id	Inspect 21 categorized clauses, view 'What You Need to Pay/Do/Avoid', ask questions in Contract Chat, download PDF.
8. Historical Logs	/history	Browse analyzed agreements, search past summaries, download PDF reports, delete outdated records.

9. Offline Hackathon Demonstration Runbook

Step-by-step checklist to execute an infallible live jury demo without internet connection:

- **Step 1: One-Click Startup:** Double-click Launch_ContractAI.bat on Desktop. Backend launches on port 8000, Vite dev launches on port 5173, and default browser opens http://localhost:5173.
- **Step 2: Authenticate Gate:** Sign in using student@university.edu and student123. Demonstrate that directly visiting /analyzer or /dashboard without credentials triggers route protection.
- **Step 3: Load Demo Records:** Navigate to Dashboard and click the green **Load Demo Data** button. Demonstrate 20 contacts populated across Teachers, College, Friends, Internship, Placement, Hackathon, and Professional.
- **Step 4: Smart Insights Pitch:** Switch to **Smart Insights** and **Risk Detection** tabs. Point out duplicate detection, inactive advisor reminders, and network health scoring.
- **Step 5: Live Contract Audit:** Navigate to **Contract AI** → Choose **Load Sample Internship Agreement** → Click **Analyze Agreement**. Point out the instant extraction of 30-day notice, INR 25,000 fine, and IP assignment clauses.
- **Step 6: Interactive Chat & Report:** Ask the Contract Chat: *'What is the notice period?'* → Show verbatim citations → Click **Download Report** to obtain the finalized document.

10. Future Roadmap & Enterprise Scaling

- **Phase 1: Multi-Language NLP:** Support Hindi, Marathi, Tamil, and regional Indian languages for rural student lease and loan contracts.
- **Phase 2: OCR Image Parsing:** Integrate Tesseract / PaddleOCR to parse mobile photos and scanned photocopies of paper agreements.

- **Phase 3: Calendar & Email Sync:** Automated Google Calendar / Outlook integration to sync notice period countdowns and recruiter follow-ups.
- **Phase 4: Multi-Student Collab:** Shared group workspaces for hackathon teams and university project cohorts to audit group NDAs.